Part 1: Visual Design Principles

Visual design principles guide how elements are arranged to make an interface balanced, organized, and easy to use[2].

- **Hierarchy & Scale:**
  - Objects are sized based on importance. The larger the object, the more important it is[3333].
  - Scale helps balance elements based on their relevance[4].
- **Emphasis:**
  - This is how you grab the user's attention[5].
  - **Techniques:** Rescaling (making things bigger), positioning (center vs. peripheral), highlighting with color/texture, using bold fonts, capitalizing text, or using arrows[6666].
- **Proportion:**
  - Object size and color share are decided based on values and importance[7].
- **Negative Space (White Space):**
  - This is the empty space surrounding elements[8].
  - Proper spacing prevents the interface from looking cluttered or flooding the user with information[9].
  - It improves readability and draws focus to important elements[10101010].
- **Color and Contrast:**
  - Used to make specific elements "pop" and stand out[11].
  - **Warm colors** stand out against dark backgrounds; **Cool colors** stand out against light backgrounds[12].
- **Repetition (Consistency):**
  - Using the same colors, logos, and fonts throughout the product creates brand identity[13131313].
  - Repetition builds "muscle memory," helping users learn the app faster[14].
- **Balance and Alignment:**
  - Alignment creates structure, telling users where to look and how to interact[15151515].
  - Symmetric distribution of elements makes the design aesthetically pleasing[16].
- **Movement & Reading Patterns:**
  - Movement tracks how a user scans content[17]. Users typically scan in specific patterns:

        - **Z-Shaped:** Top left to bottom right (common for scanning)[18].
        - **F-Shaped:** Reading the top line, then scanning down the left side (like Google results)[19].
        - **Layered Cake:** Reading only headings and titles, ignoring the rest[20].
- **Unity & Variety:**
    - **Unity:** Presenting information in a cohesive, logical way using harmonious colors and similar shapes[21212121].
    - **Variety:** Breaking monotony to keep the interface interesting[22].
- **Proximity:**
    - Group related items together to show they are connected; space unrelated items apart[23].

Part 2: Interaction Design (IxD)

The goal of IxD is to create products that help users achieve their objectives efficiently[24].

The 5 Dimensions of Interaction Design:

1. **Words:** Should be meaningful and simple (e.g., button labels) so they don't overwhelm the user[25].
2. **Visual Representations:** Icons, typography, and images that supplement words[26].
3. **Physical Objects/Space:** The device the user interacts with (e.g., mouse, touchscreen)[27].
4. **Time:** Media that changes over time (animations, video, sound) and the ability for users to resume progress later[28282828].
5. **Behavior:** The mechanism of the product—how users operate it and how the product reacts (feedback/emotion)[29292929].

Part 3: Prototyping Tools

Two major tools were highlighted for creating digital designs:

1. Sketch

- A vector graphics editor used for wireframing and prototyping[30].
- **Key Features:** Infinite design canvas, flexible Artboards, simple resizing tools, and mathematical operators to speed up design[31].

2. Proto.io

- A web-based, **no-code** tool for high-fidelity interactive prototypes[32].
- **Key Features:** Drag-and-drop interface, huge library of UI components (iOS/Android), animation features, and integration with tools like Figma and Photoshop[33333333].

Part 4: Usability Testing

What is Usability?

- **Definition:** The absence of frustration when using a product[34].
- A product is usable if it is: **Useful, Efficient, Effective, Satisfying, Learnable, and Accessible**[35].
    - **Usefulness:** Does it actually achieve the user's goals? If not, they won't use it, even if it's free[36].
    - **Efficiency:** How quickly can the goal be accomplished?[37].
    - **Effectiveness:** Does the product behave the way users expect?[38].
    - **Learnability:** Can a user operate it after training (or with no training)?[39].
    - **Accessibility:** Can people with disabilities use it? (This benefits *everyone*) [40].

Why Products Fail at Usability

1. **Focus on Machine:** Developers focus on the system rather than the human end-user[41].
2. **Changing Audiences:** The gap between developers (tech-savvy) and users (less patience/tech knowledge) is widening[42424242].
3. **Silos:** Specialized teams (UI, Help, Technical) often don't integrate their work effectively[43].
4. **Design vs. Implementation Mismatch:** "Design" is communication; "Implementation" is how it works. These often conflict[44].

Types of Usability Tests

Testing should happen throughout the product lifecycle[45454545].

1. **Exploratory (Formative) Test:**
    - **When:** Early stage (pre-design or preliminary design)[46].
    - **Goal:** Check effectiveness of concepts and basic layout[47474747].
    - **Method:** Informal collaboration between moderator and user; "thinking aloud" is encouraged[48].
2. **Assessment (Summative) Test:**
    - **When:** Midway through development[49].
    - **Goal:** Evaluate lower-level operations/features[50].
    - **Method:** Users perform specific tasks; less interaction from the moderator; quantitative data is collected[51].
3. **Validation (Verification) Test:**
    - **When:** Late stage, close to release[52525252].
    - **Goal:** Confirm the product meets specific benchmarks or standards[53535353].

    - **Method:** Strict experimental rigor; users perform tasks alone; focus on error rates and time measures[545454545454545454].

4. **Comparison Test:**
    - **When:** Any time[55].
    - **Goal:** Compare two designs (A vs. B) or compare your product against a competitor[56].

The Testing Process

1. **Develop Test Plan:** The blueprint covering goals, research questions, and participant characteristics[575757575757575757].
2. **Set Up Environment:**
    - **Lab:** Good for control and close observation, but artificial[58585858].
    - **Portable Lab:** Bring equipment to the user; captures context but logistically complex[59].
    - **User's Site:** Real environment, but hard to arrange[60].
3. **Find Participants:** Must represent the actual target audience. Use "Personas" to categorize them[61616161].
4. **Prepare Materials:** Scripts, questionnaires (pre/post-test), and task scenarios[62626262].
5. **Conduct Session:** Run the test[63].
6. **Debrief:** Talk to participants and observers after the test[64].
7. **Analyze Data:** Review quantitative and qualitative measures[65].
8. **Report:** Share findings and recommendations[66].